

Performance Report

Evaluation

Installation	Equipment	Type	No.
APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356

Readings Included

	Date	Voyage / title	Remarks
1	2025-09-26	04FLAE1MA	ENROUTE FROM AFRICA TO SINGAPORE

Main indications (all Readings)

Reading	1
Description	Most recent
Fuel oil saving potential	●
Light running	●
Fuel index ECU	●
Firing press. AVG	●
Compression press. AVG	●
Scav. air press. RECEIVER	●
Exh. gas temp. TC in AVG	●
Exh. gas temp. TC out AVG	●
TC speed AVG	●
Spec. FO consumption	●

Indications (for Reading 1)

Description	Indication	Comments / recommendations
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Optimization potential

Quantification						
Influenced by	Fuel oil equiv.	Thermal load	Proposed actions to achieve the saving / reduction			
Firing pressure by injection	0.0 g/kWh	0 °C				
Firing pressure by compression ratio	0.0 g/kWh	0 °C				
Firing pressure by scavenge air pressure	0.3 g/kWh	12 °C	Increase scav. air press. to raise firing press. by 1 bar. Or advance inj. begin to compensate.			
			Target firing press.: 146.8 bar (ISO corr., REF.) or 147 bar (meas., AVG.). Check OEM limits pmax - pcomp: 35 bar.			
ENGINE ROOM ventilation	0.0 g/kWh	0 °C				
Suction press. air TC in	0.0 g/kWh	0 °C				
Press. drop accross SAC	0.0 g/kWh	0 °C				
Scav. air : water temp. SAC in	0.0 g/kWh	0 °C				
Scav. air : temp. RECEIVER	0.2 g/kWh	1 °C	Increase water flow, clean SAC water- and/or air side to reduce scav. air temp. by 2 °C to 37 °C.*			
Press. drop accross ENGINE						
Exh. gas press. TC out	0.0 g/kWh	0 °C				
Light running	0.0 g/kWh	-				
Fuel oil visc. ENGINE in	0.0 g/kWh	0 °C				
	Fuel oil equiv.	Thermal load	Cyl. lub. oil			
TOTAL (specific)	0.5 g/kWh	-				
TOTAL (of current)	0.3 %	-		CO2 (emission)	SOx (emission)	
TOTAL optimization potential	0.29 t/d	13 °C		0.9 t/d	3 kg/d	
	Cost					
Fuel oil price						
TOTAL saving potential						

Consumption

Measured / calculated values			
Indication	Fuel oil		Cylinder lub oil
Specific consumption reference	165.7 g/kWh		0.70 g/kWh
Specific consumption ISO / MEAS	168.4 g/kWh		0.70 g/kWh
Consumption per hour day *	4659 kg/h	111.8 t/d	18.5 kg/h
Consumption nautical *	233 kg/nmi		
* Measured values, uncorrected			

Injection timing and compression ratio adjustment

Cylinders												
Settings		1	2	3	4	5	6	7	8	9	10	
Proposed adjustments to reach target firing press., positive value means increase of pressure:												
Compr. ratio offset [-]		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
pmax offset [bar]		-0.1	-0.1	0.0	0.0	0.0	-0.1	0.0	0.0	0.0	0.0	-0.1
Common compr. ratio offset [-]	-0.2											
Common pmax offset [bar]	1.1											

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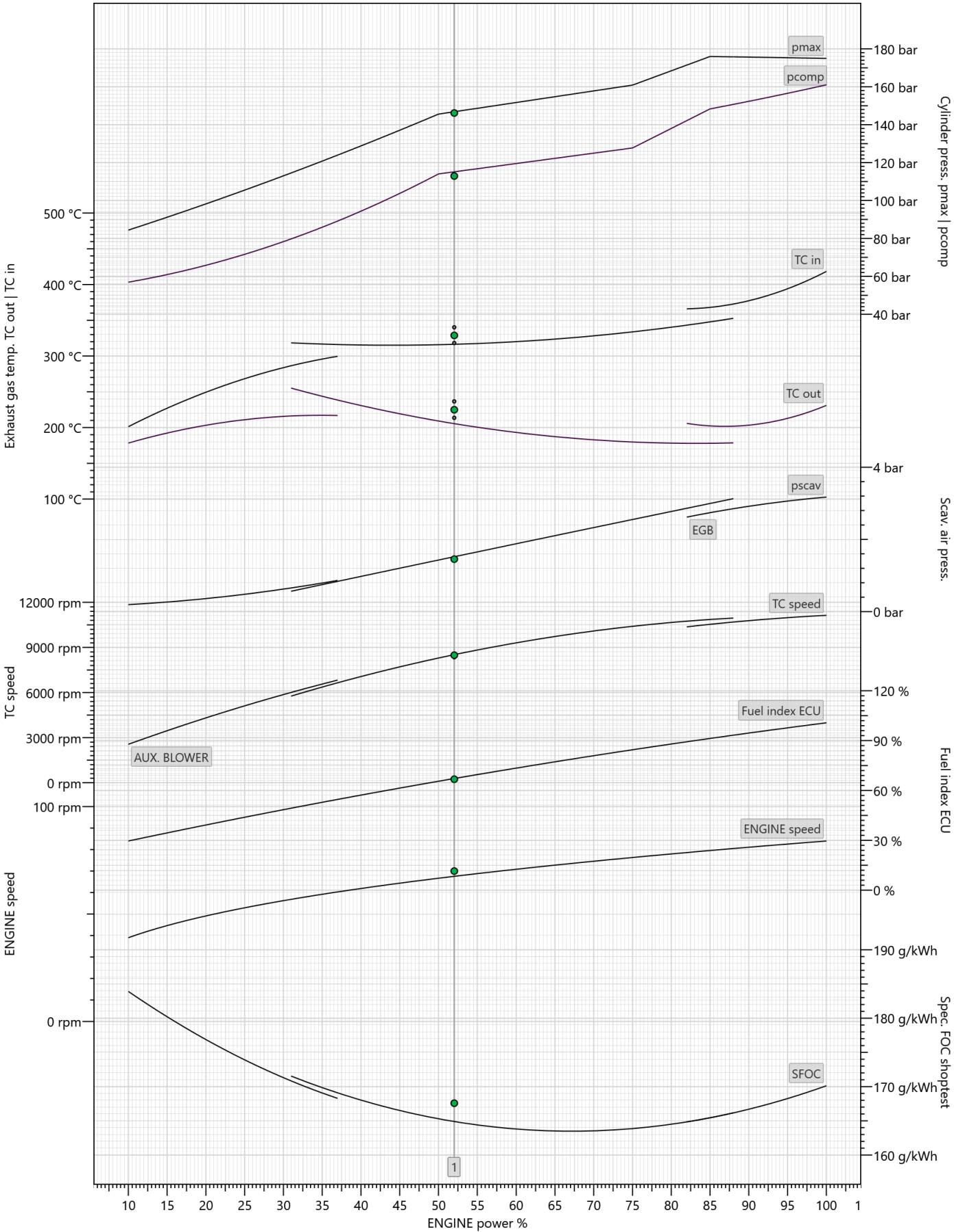
Input plausibility

Value	Comment
Compressor evaluation	
Exhaust gas pressure Manifold	
PTI	
PCO	
TTI vs TTO	
TCO	
Minimum required data	
Minimum required data II	
Missing data	
Missing data II	

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Performance Curves

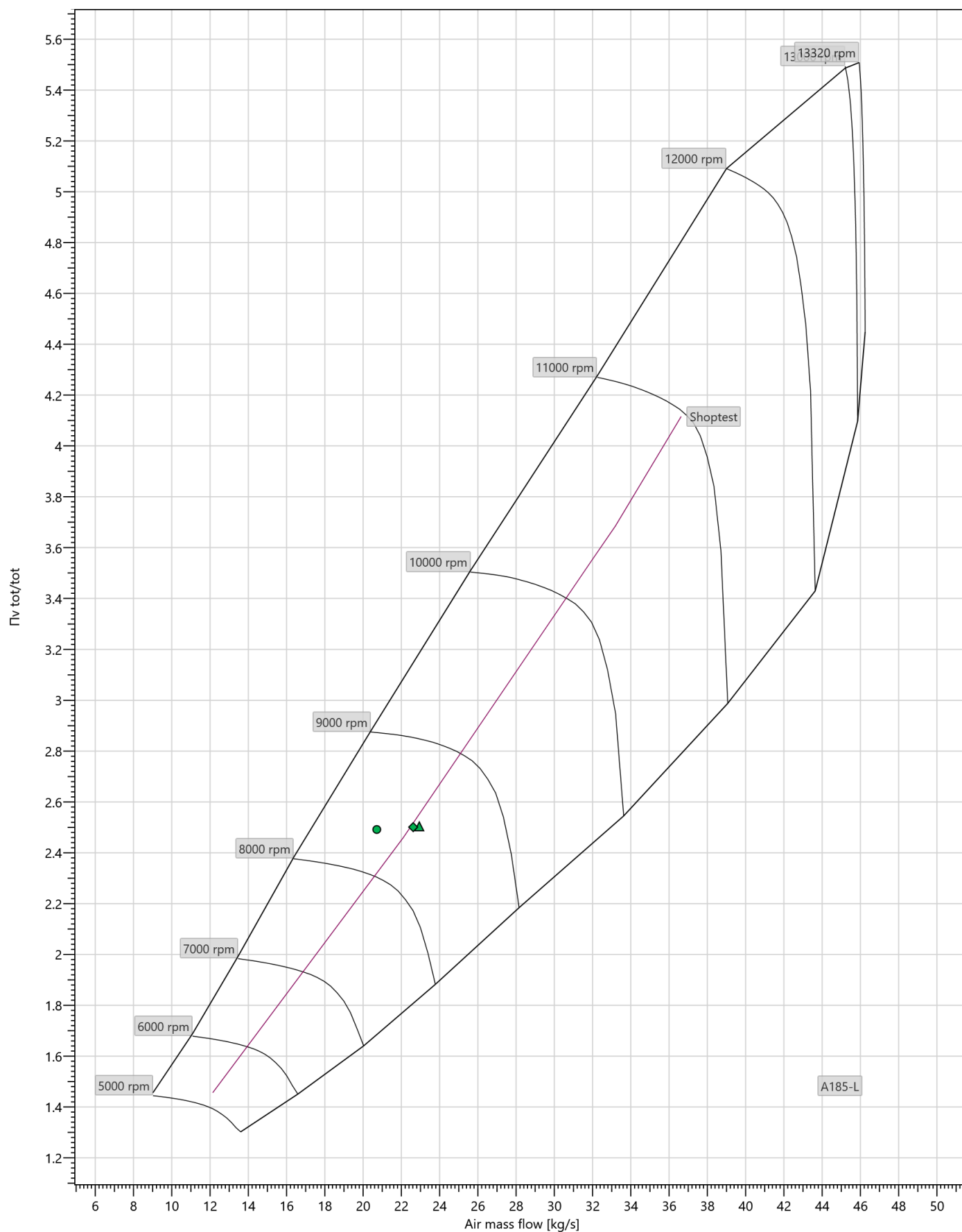
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COMPRESSOR map

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APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356

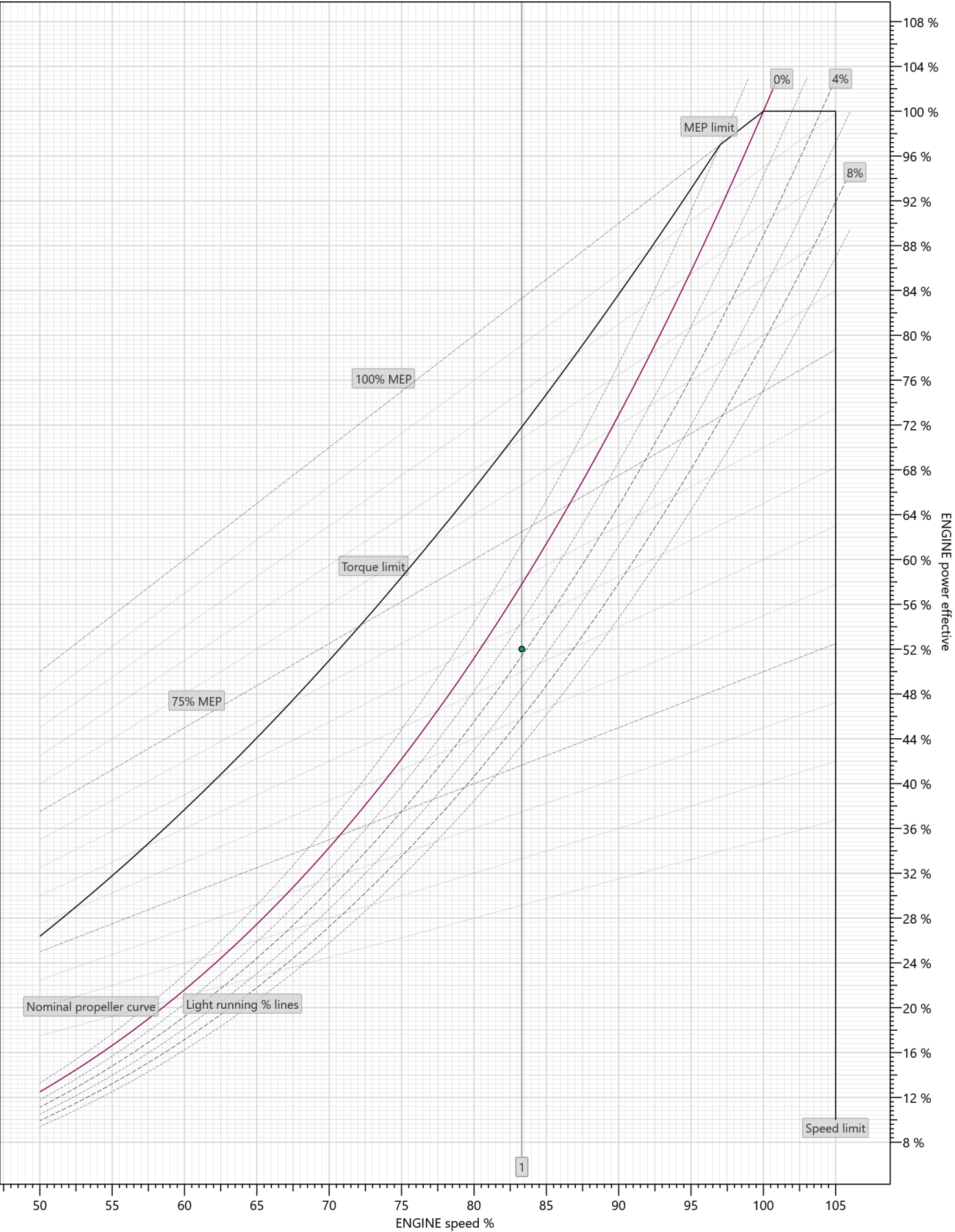


1 2025-09-26 04FLAE1MA		
● TC 1	▲ TC 2	◆ TC 3

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Load Diagram

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Reading

Installation		Equipment				Type				No.																				
APL HOUSTON				MAIN ENGINE				10S90ME-C				DML0105356																		
General	Date and time of recording		9/26/2025 15 : 35		Sea water temp.		27.0 °C		SHIP SOG STW		19.00 20.00 kn		Voy. / title		04FLAE1MA															
	ENGINE state		stable		Ambient air temp.		29.0 °C		Draft fwd aft		10.50 11.40 m		Record by		2/E															
	ENGINE running hours		69064 h		Engine room air temp.		30.0 °C		Draft mid Trim		10.80 0.90 m		Ackn. by		C/E															
	Ambient humidity		rel 45.0 % abs 11.4 g/kg		Barom. pr. Eng. room		1.000 1.018 barA		Remarks		ENROUTE FROM AFRICA TO SINGAPORE																			
Power / Speed			CALCULATED		MEAS		ESTIMATED 24.3(E264X126)																							
			XPERT		TORSIOM.		MOP		by FI*rpm		by pscav		by TC rpm		by MEP		by rpm													
	ENGINE power estimated		kW		26560 26441		26556		26286		25925		26760		25897															
			%		52.0		52.0																							
			MCR		ENGINE		TORSIOMETER		MARGINS		MECH. EFFICIENCY																			
Electronic ctrl.	Running mode		Economy		ISO CORRECTED		MEASURED																							
	Governor mode		RPM		REF.		CALC.		ALL																					
	Fuel index ECU		%		67.3 66.8		64.0																							
	Applied fuel quality offset		%				0.0																							
Cylinder pressure	SOURCE: MAN CoCoS		ISO CORRECTED		MEASURED										SETPOINT															
	QUALITY Imported		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10		MOP	
	Firing press. pmax		barG		146.8 146.2		146.2		146.2		146.1		146.2		146.1		146.2		146.2		146.2		146.2		146.2		146.2			
	pmax deviation		bar				ALL		0.0		0.0		-0.1		0.0		-0.1		0.0		0.0		0.0		0.0		0.0		0.0	
	pmax offset		bar				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Compression press.		barG		115.1 112.9		112.7		112.7		112.7		112.8		112.7		112.7		112.8		112.8		112.8		112.8		112.7			
	pcomp deviation		bar				ALL		0.0		0.0		0.0		0.1		0.0		0.0		0.1		0.1		0.1		0.0		0.0	
	pcomp / pscav offset		-				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Mean indicated press.		barG		11.67		11.40		11.40		11.40		11.40		11.40		11.40		11.40		11.40		11.40		11.40		11.40			
	MIP deviation		bar				0.00		0.00		0.00		0.00		0.00		0.00		0.00		0.00		0.00		0.00		0.00		0.00	
	Fuel offset high		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Fuel offset low		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
	Power indicated		kW				SUM:		27575		2758		2758		2758		2758		2758		2758		2758		2758		2758			
	Mean effective press. (MEP)		bar		10.98		10.71		60.9 %		of MCR																			
pmax-pcomp pcomp / pscav		-		31.7 46.2		33.4 46.4												pcomp / pscav		46.6										
Turbobcharger			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG. / ENG.		TC 1		TC 2		TC 3																	
	PV (tot-tot)								2.49		2.50		2.50																	
	TC speed		rpm		8510 8480		8552		8415		8657		8584																	
	TC speed 298		rpm		8376 8414		8442		8306		8545		8473																	
	Turbobcharger efficiency																													
	Surge margin																													
	Air mass flow		kg/s		68.3 65.2		64.3		20.1		22.3		21.9																	
	Break spec.air cons.		kg/kWh		8.7																									
	Exhaust gas mass flow		kg/s		68.8 65.9		64.9		20.3		22.5		22.2																	
Scavenge air			ISO CORRECTED		MEASURED																									
	AUX. BLOWER operation		off		REF.		CALC.		AVG.		TC 1		TC 2		TC 3															
	Suction press.		mmWG		27		27		27		27		27		27															
	Press. drop accr. SAC		mmWG		156		151		150		152		152																	
	Scav. air press. RECEIVER		barG		1.51 1.45		1.45																							
	Scav.air press.setp. MOP		barG																											
	Air temp. COMPR. in		°C				31.0		31.0		31.0		31.0																	
	Press. COMPR. outlet *		barG		1.51		1.45		1.45		1.45		1.45																	
	Scav. air temp. COMPR. out		°C		138		138		138		138		138																	
	Scav. air temp. SAC out		°C				38.0		40.0		40.0																			
	Scav. air temp. RECEIVER		°C		37.0		39.0																							
	Cool. water temp. SAC in		°C		33		32.0		32.0		32.0		32.0																	
	Cool. water temp. SAC out		°C				40.0		40.0		40.0		40.0																	
	*measured at flange																													
Exhaust gas			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10			
	Exh. gas temp. CYL out		°C		285 282		295		285		303		310		320		297		282		276		295		289		295			
	temp. deviation		°C				ALL		-10		8		15		25		2		-13		-19		0		-6		0			
	Exh. vlv. open offset		°CA				MOP SETTING:																							
	Exh. gas press. MANIFOLD		barG		1.37				TC 1		TC 2		TC 3																	
	Exh. gas press. TC in *		barG		1.34		1.20		1.20		1.20		1.20																	
	Exh. gas press. TC out		mmWG		81		74		73		74		74																	
	Exh. gas temp. TC in		°C		318 329		342		353		341		331																	
	Exh. gas temp. TC out		°C		208 225		236		248		236		225																	
	Exh.wastegate (EGB) pos. area %				0 0 2 1		1																							
	*measured at flange																													

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Installation		Equipment		Type		No.	
APL HOUSTON		MAIN ENGINE		10S90ME-C		DML0105356	
Fuel oil		ISO CORRECTED	MEAS	Fuel oil properties:	ANALYS.	ESTIMATE	Fuel oil system:
	Absolute consumption	m ³ /h	REF. CALC.	5.020	Viscosity	cSt 132 @ 50 °C	Viscosity ENGINE in
	Specific FOC	g/kWh	165.7	168.4	175.4	Sulfur content	% 0.50 0.50
	Light running correction	g/kWh	0.0			Water content	% 0.20 0.20
	Thrust bearing loss	g/kWh	-0.8			LCV	kJ/kg 41380 41380
	Specific FOC shoptest	g/kWh	164.9	167.6		Density	kg/m ³ 976.0 @ 15 °C
					Source of analysis	Laboratory	Price
Cyl. lub.		REFERENCE	MEAS	Cyl. lubr. system (Alpha):	REF	SET**	Cyl. lub. oil:
	Absolute consumption	l/h	20.0	Target feed rate@MCR	g/kWh 0.60	0.70	BN
	Spec. lub oil consumpt. *	g/kWh	0.70	Target f.r.@current load	g/kWh 0.70	SET based	Density
	* Effective feed rate			** Acc'to OEM operation manuals and/or drainoil analysis			kg/m ³ 940.0 @ 15 °C
Cylinder condition		REFERENCE	MEASURED				
		REF.	AVG.	ENGINE / CYL 1			
	Cool. water temp. CYL in	°C 75	79.0	79	CYL 2	CYL 3	CYL 4
	Cool. water temp. CYL out	°C 90	87.0	87	87	88	88
	Cooling water press. CYL in	bar	5.0	5.0	5.0	5.0	5.0
	Coolant temp. PISTON in	°C	45	45	45	45	45
	Coolant temp. PISTON out	°C	56	55	55	56	56
	Scav. air temp. PIST. UNDERS.	°C	48.7	47	45	50	51
					50	48	48
					49	49	50
Crankcase			MEASURED				
			AVG. / ENG.	BRG 1	BRG 2	BRG 3	BRG 4
	MAIN BEARING temp.	°C	54.3	53.0	56.0	56.0	55.0
	Temp. deviation	°C		-1	2	2	1
				1	1	-1	-3
				-1	-1	-1	-1
				3	2	3	-3
Liner wall			MEASURED				
				CYL 1	CYL 2	CYL 3	CYL 4
	LINER WALL temp. aft	°C					
	LINER WALL temp. fp	°C		126	134	139	142
	LINER WALL temp. fwd	°C					
	LINER WALL temp. exh	°C					
	LINER WALL temp. AVG	°C	132	121	132	139	132
TC/SAC media			MEASURED				
			AVG.	TC 1	TC 2	TC 3	
	Water press. SAC in	bar	2.9	2.9	2.9	2.9	
	Lub oil temp. TC in	°C	45	45	45	45	
Engine media			MEAS				
	Lub oil temp ENGINE in	°C	45.0	Fuel oil press. ENGINE in	bar	7.50	
	Lub oil temp. THRUST SEGM.	°C	58.0	Fuel oil press. ENGINE out	bar	7.20	
	Lub oil press. SYSTEM OIL	bar	2.40	Fuel oil press. FILTER in	bar	7.5	
	Lub oil press. COOLING OIL	bar	2.30	Fuel oil press. FILTER out	bar	7.2	
	Lub oil press. TURB. OIL	bar	2.20	Hydraulic oil press. FILTER in	bar	2.4	
	Lub oil temp. TURB. OIL in	°C	45.00	Hydraulic oil press. FILTER out	bar	2.1	
	Lub oil temp. TURB. OIL out	°C	62.33	Hydraulic oil press. MAIN	bar	233.0	
	Axial vibration	mm	0.71	Sealing air press.	bar	7.5	PUMP 2 PUMP 3 PUMP 4 PUMP 5
				Hydr. oil PUMP swash plate pos.	%	60.0	50.0 60.0
Plant			MEAS				
	Exh. gas differential press. ECONOMIZER	mmWG	30.00				